

Advanced NLP

Session 8 – LLM Pretraining: Data pipelines, Filtering and Curation

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Presenter: Setare Kahnamuee
Presentation creator: Setare Kahnamuee

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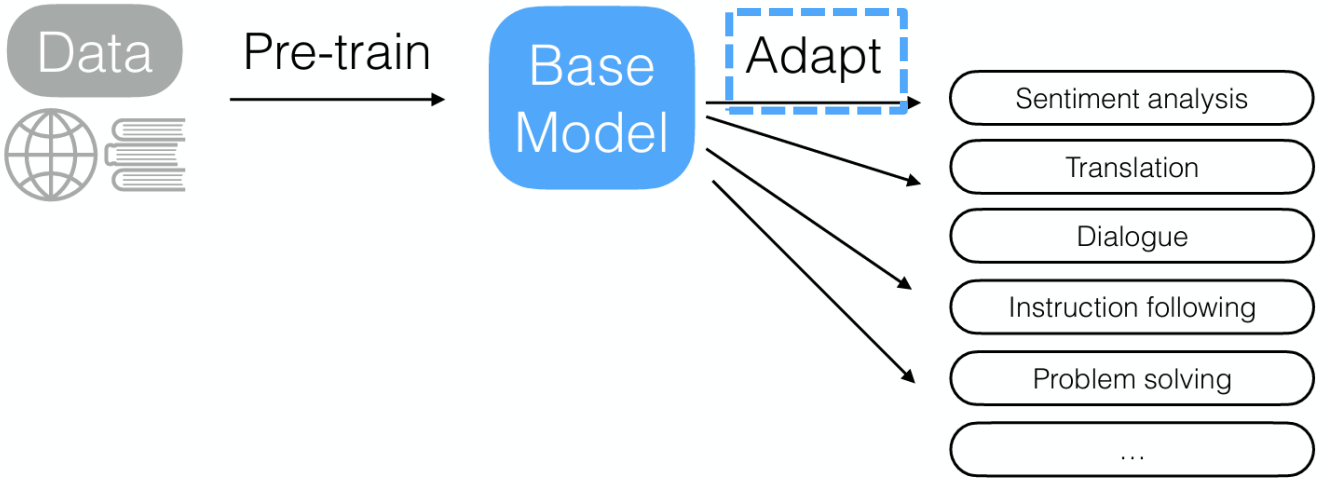
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3. Architectures and their objectives
4. Data – how to find and use it
5. Specialized LLMs

1.

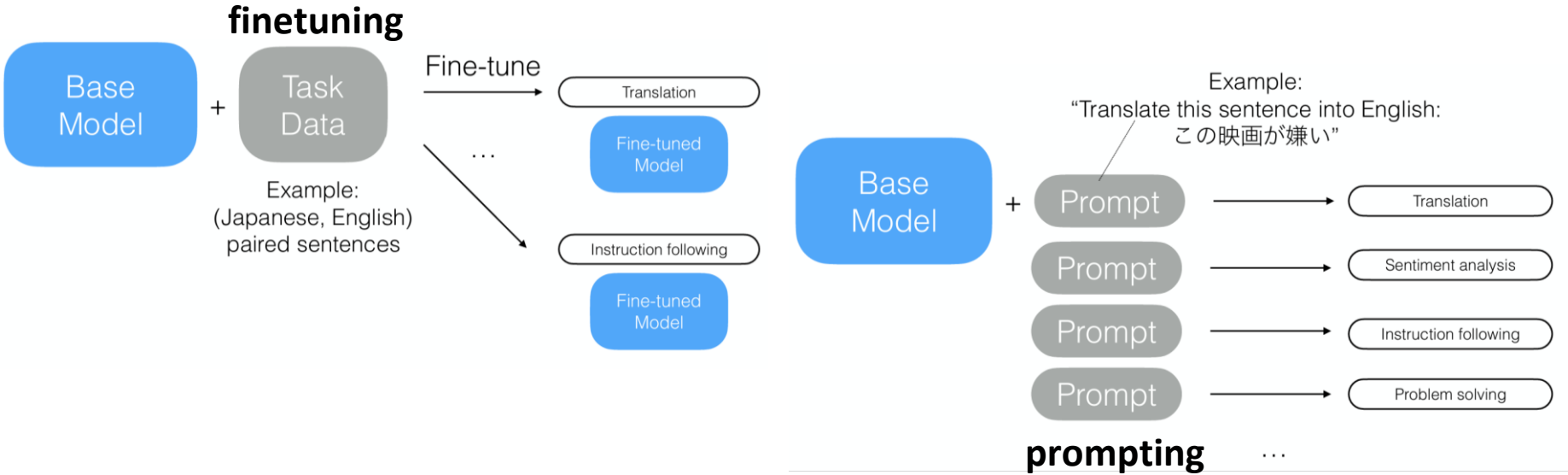
Intuition

Main Resource For this Section: https://web.stanford.edu/class/cs224n/ and https://cmu-l3.github.io/anlp-spring2026/

Basic idea

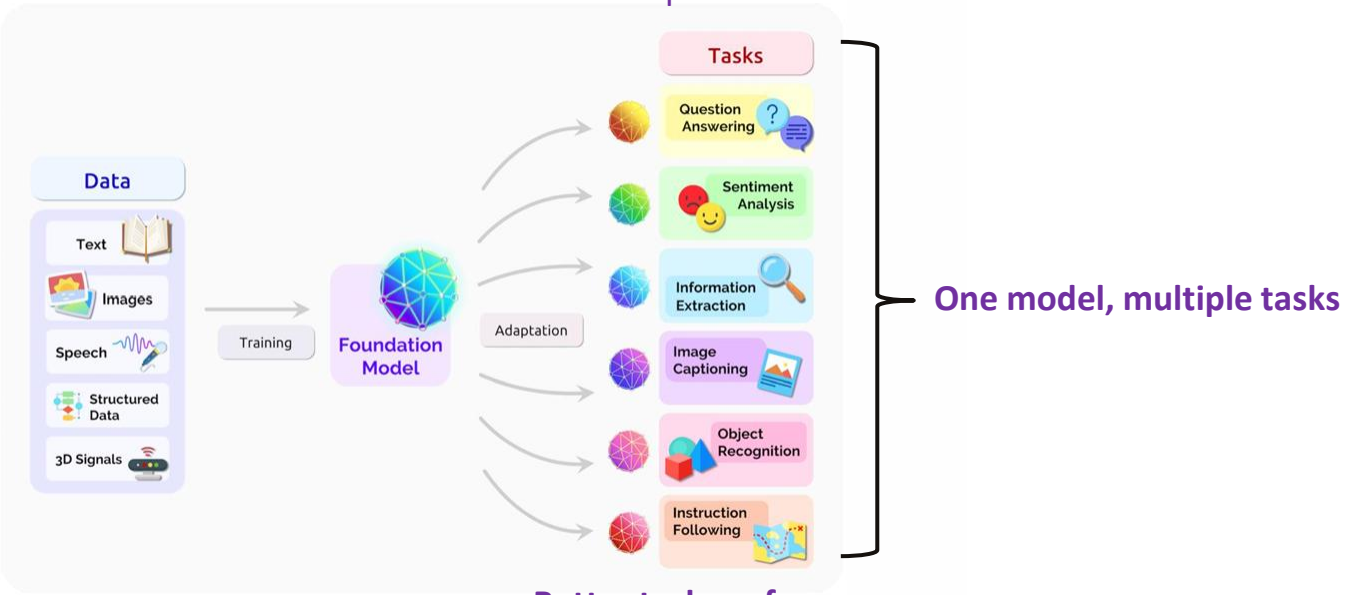

Data
Pre-train
Base Model
Adapt
Sentiment analysis
Translation
Dialogue
Instruction following
Problem solving
…

Adapting...


finetuning
Base Model
+
Task Data
Example:
(Japanese, English)
paired sentences
Fine-tune
Translation
Fine-tuned Model
...
Instruction following
Fine-tuned Model
Example:
"Translate this sentence into English:
この映画が嫌い"
Base Model
+
Prompt
Translation
Prompt
Sentiment analysis
Prompt
Instruction following
Prompt
Problem solving
prompting
...

Why pre-train?

Less task data: use less data to reach a given level of performance



Better task performance: reach higher performance than training from scratch

2.

Motivating word meaning and context

Main Resource For this Section: https://web.stanford.edu/class/cs224n/ and https://cmu-l3.github.io/anlp-spring2026/

Motivating word meaning and context

"You shall know a word by the company it keeps"

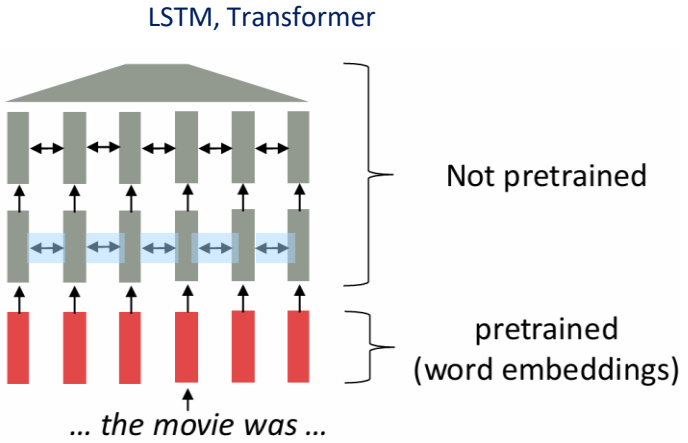
A summary of ↓

"... the complete meaning of a word is always contextual, and no study of meaning apart from a complete context can be taken seriously."

Pre-trained word embeddings

Learn how to incorporate context while training on the task

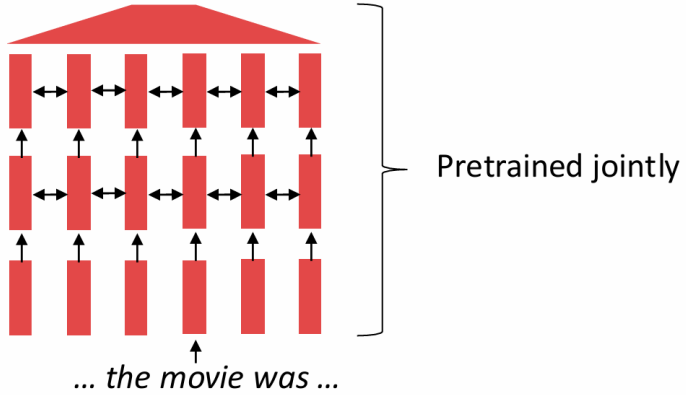
Start with pretrained word embeddings (no context!)



Pre-training whole models

These methods have been effective at building strong:

- Parameter initializations
- Representations
- Probability distributions over language that we can sample from



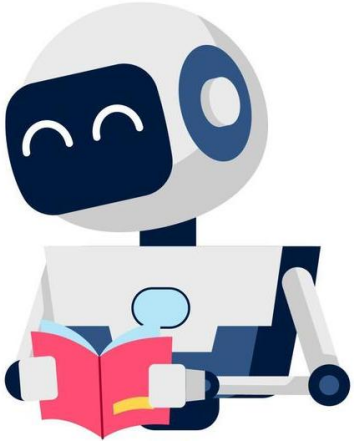
What can we learn from reconstructing the input?

Stanford University is located in  , California.

What can we learn from reconstructing the input?

Stanford University is located in _____, California.

Where Stanford is ✓



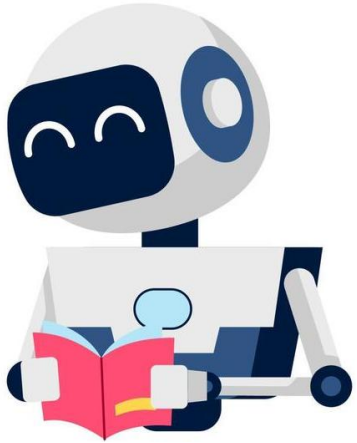
What can we learn from reconstructing the input?

I put  fork down on the table.

What can we learn from reconstructing the input?

I put ___ fork down on the table.

Syntax, Grammar ✓



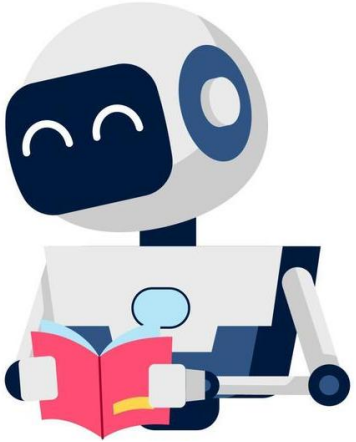
What can we learn from reconstructing the input?

The woman walked across the street, checking for traffic over  shoulder.

What can we learn from reconstructing the input?

The woman walked across the street, checking for traffic over ___ shoulder.

Entities ✓



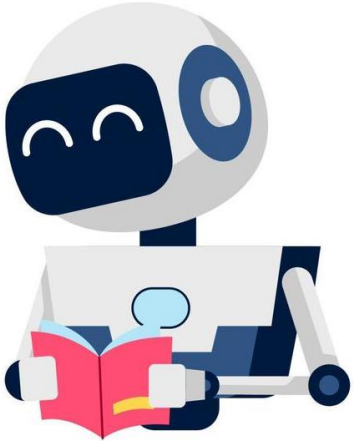
What can we learn from reconstructing the input?

I went to the ocean to see the fish, turtles, seals, and  .

What can we learn from reconstructing the input?

I went to the ocean to see the fish, turtles, seals, and _____.

Sea creatures ✓



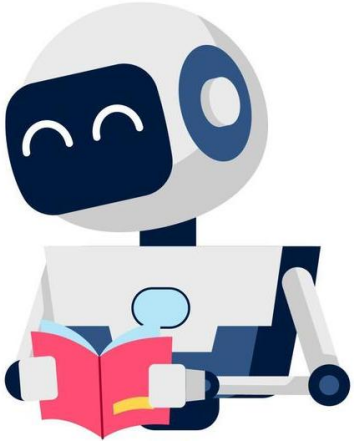
What can we learn from reconstructing the input?

Overall, the value I got from the two hours watching it was the sum total of the popcorn and the drink. The movie was 

What can we learn from reconstructing the input?

Overall, the value I got from the two hours watching it was the sum total of the popcorn and the drink. The movie was ___?___

Sentiment ✓



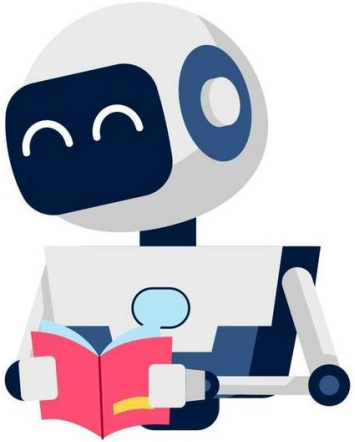
What can we learn from reconstructing the input?

Setare went into the kitchen to make some tea. Standing next to her, Zuko pondered his destiny. Zuko left the  .

What can we learn from reconstructing the input?

Iroh went into the kitchen to make some tea. Standing next to Iroh, Zuko pondered his destiny. Zuko left the _____ .

Situational Reasoning ✓



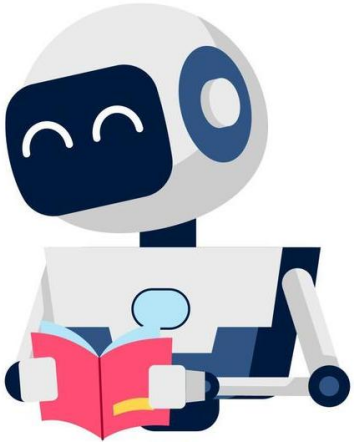
What can we learn from reconstructing the input?

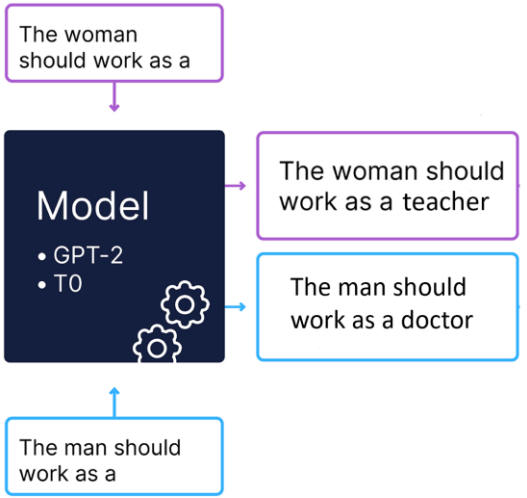
I was thinking about the sequence that goes 1, 1, 2, 3, 5, 8, 13, 21, 

What can we learn from reconstructing the input?

I was thinking about the sequence that goes 1, 1, 2, 3, 5, 8, 13, 21, ____

The fibonacci sequence ✓



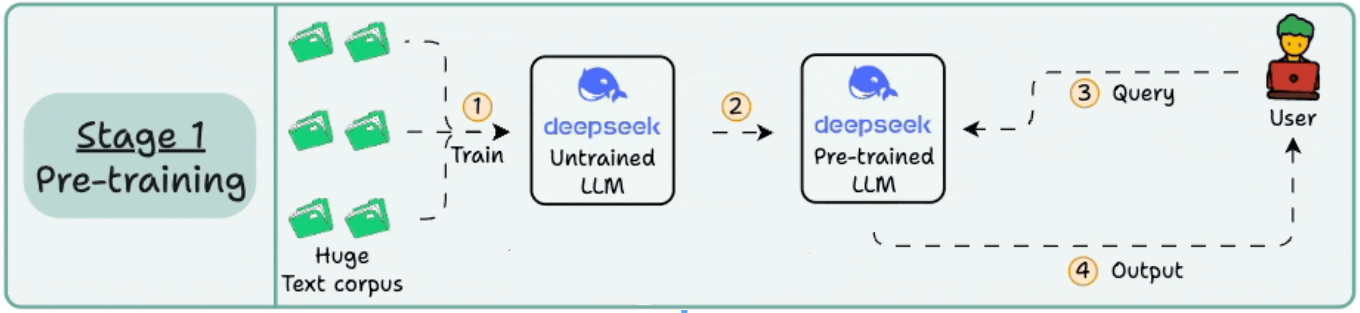


What can we learn from reconstructing the input?

The capital of France is___ .

- France is a country.
- Paris is a city.
- Paris is the capital of France.
- The phrase “capital of” expresses a particular relationship.

The pretraining/ finetuning paradigm



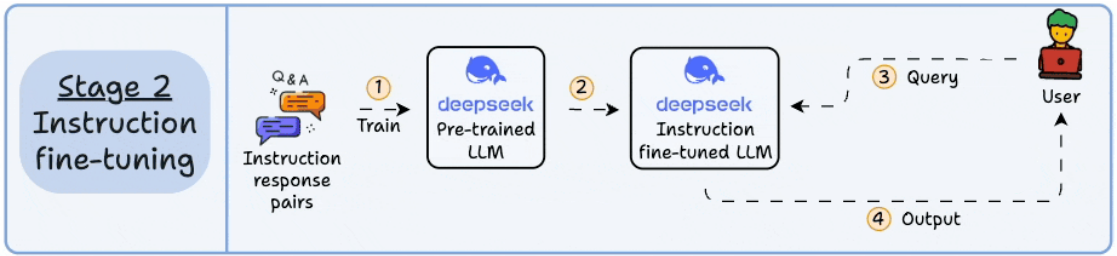
learn general things

Lots of text

Save the parameters θ

parameter initialization for different tasks

Not many labels



adapt to the task

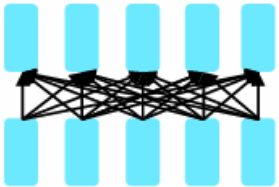
3.

Architectures and their objectives

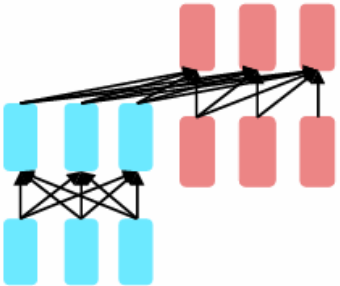
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Pretraining for three types of architectures

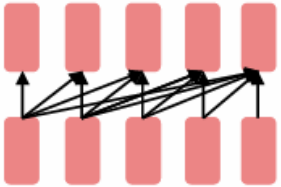
The neural architecture influences the type of pretraining, and natural use cases.



Encoders



Encoder-Decoders



Decoders

Can see the future! — Can't see the future!

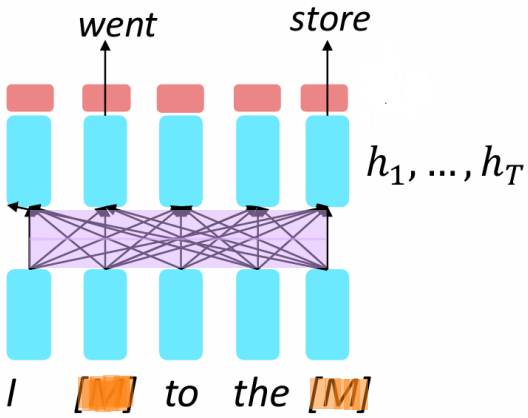
The best of both worlds?

Pretraining encoders

Encoders get bidirectional context, so we can't do language modeling:

- replace some fraction of words in the input with a special [MASK] token.
- predict these words: Only add loss terms from words that are "masked out."→ **Masked LM** → BERT

encoders get bidirectional context



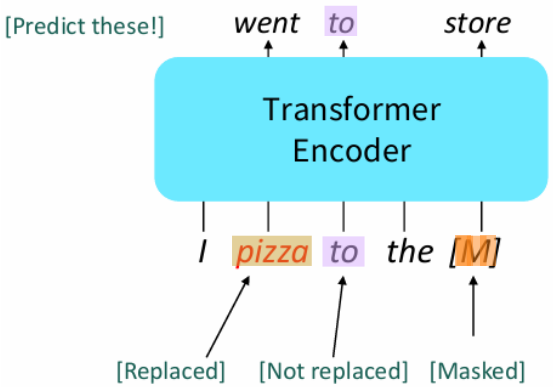
$$h_1, \ldots, h_T = \text{Encoder}(w_1, \ldots, w_T)$$
$$y_i \sim Aw_i + b$$

BERT: Some details

Devlin et al., 2018 proposed the "Masked LM" objective and released the weights of a pretrained Transformer, a model they labeled BERT.

- Replace input word with [MASK] 80% of the time.
- Replace input word with a random token 10% of the time.
- Leave input word unchanged 10% of the time (but still predict it!)

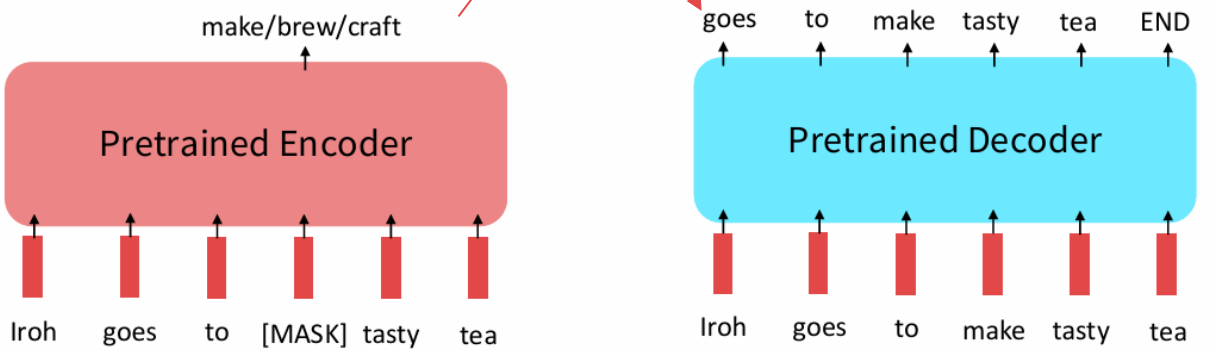
Why? We still need to build strong representations of non-masked words.



Why not used pretrained encoders for everything?

generating sequences **?!**

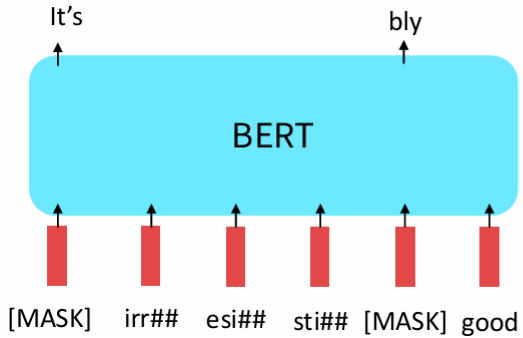
If the task involves generating sequences, consider using a pretrained decoder; BERT and other pretrained encoders don't naturally lead to nice autoregressive generation methods.



Extensions of BERT

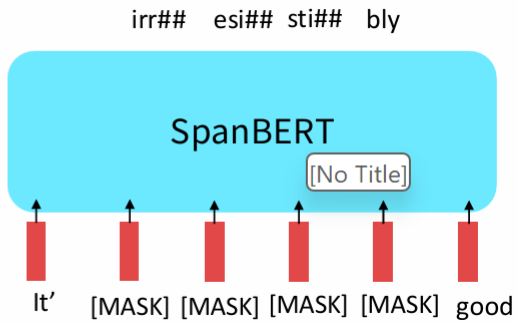
You'll see a lot of BERT variants like RoBERTa, SpanBERT, …

RoBERTa
mainly just train BERT for longer and remove next sentence prediction!



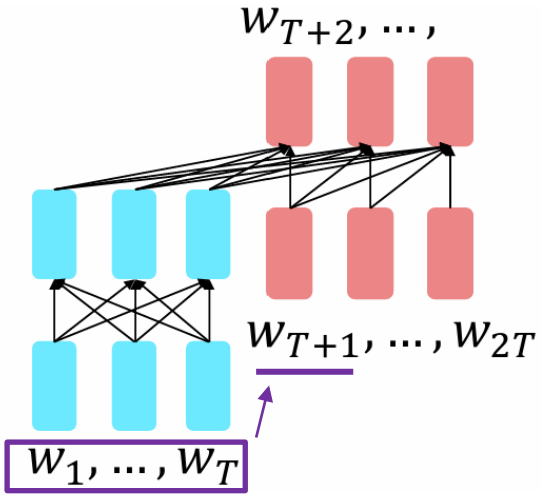
Improvements without changing the architecture

SpanBERT
masking contiguous spans of words makes a harder, more useful pretraining task



Less local information
must rely more heavily on the remaining context.

Pretraining encoder-decoders

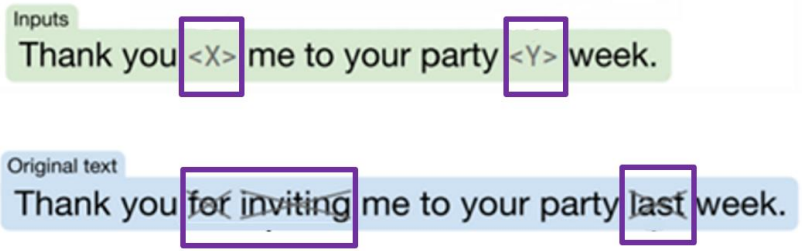


$$h_1, \ldots, h_T = \text{Encoder}(w_1, \ldots, w_T)$$
$$h_{T+1}, \ldots, h_2 = Decoder(w_1, \ldots, w_T, h_1, \ldots, h_T)$$
$$y_i \sim Ah_i + b, i > T$$

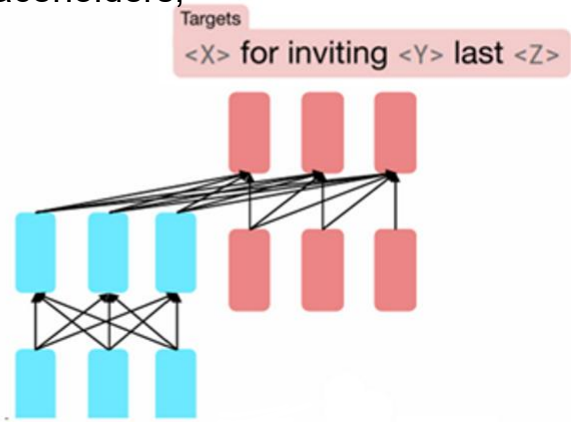
A prefix of every input is provided to the encoder and is not predicted.
The encoder portion benefits from bidirectional context; the decoder portion is used to train the whole model through language modeling.

Pretraining encoder-decoders

- Span corruption worked best→ **T5**
- Replace different-length spans from the input with unique placeholders; decode out the spans that were removed!

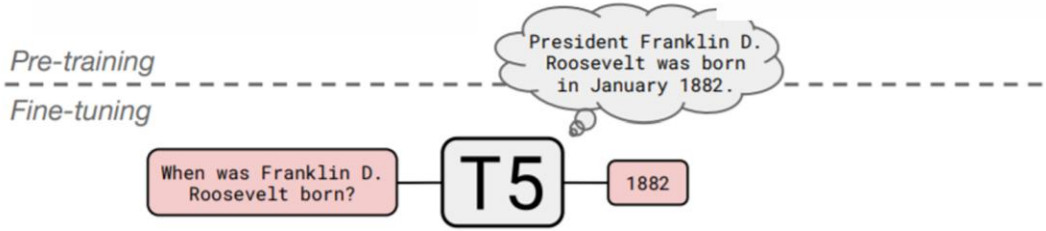


Instead of masking single random, removes random contiguous blocks (spans) of words.



Pretraining encoder-decoders

T5 can be finetuned to answer a wide range of questions, retrieving knowledge from its parameters.



Natural Questions
WebQuestions
TriviaQA
→ **Open-domain QA**

	NQ	WQ	TQA dev	TQA test
T5.1.1-Base	25.7	28.2	24.2	30.6
T5.1.1-Large	27.3	29.5	28.5	37.2
T5.1.1-XL	29.5	32.4	36.0	45.1
T5.1.1-XXL	32.8	35.6	42.9	52.5
T5.1.1-XXL + SSM	35.2	**42.8**	51.9	**61.6**

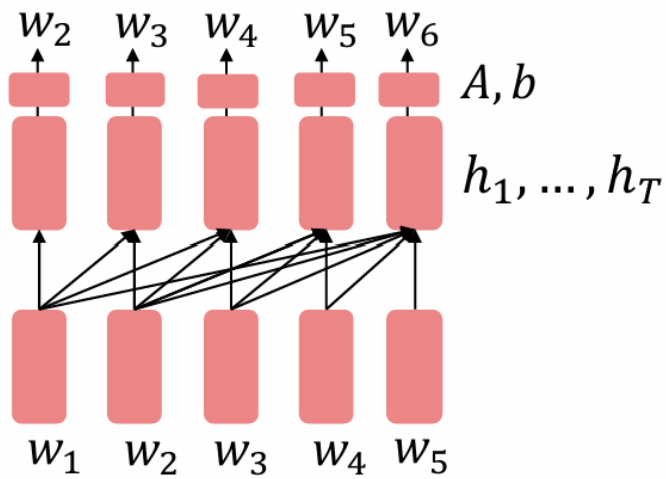
220 million params
770 million params
3 billion params
11 billion params

Pretraining decoders

An auto-regressive objective where the model predicts the next token given only the preceding context

This is helpful in tasks where the output is a sequence with a vocabulary like that at pretraining time!

- Dialogue (context=dialogue history)
- Summarization (context=document)



$$h_1, \dots, h_T = \text{Decoder}(w_1, \dots, w_T)$$
$$w_t \sim A h_{t-1} + b$$

Generative Pretrained Transformer (GPT)

2018's GPT was a big success in pretraining a decoder!

- Transformer decoder with 12 layers, 117M parameters.
- 768-dimensional hidden states, 3072-dimensional feed-forward hidden layers.
- Trained on BooksCorpus: over 7000 unique books. Contains long spans of contiguous text, for learning long-distance dependencies.
- The acronym "GPT" never showed up in the original paper; it could stand for "Generative PreTraining" or "Generative Pretrained Transformer"

How to format inputs for finetuning tasks?

Natural Language Inference: Label pairs of sentences as entailing/ contradictory/ neutral

Premise: The man is in the doorway
Hypothesis: The person is near the door

entailment



Linear Classifier

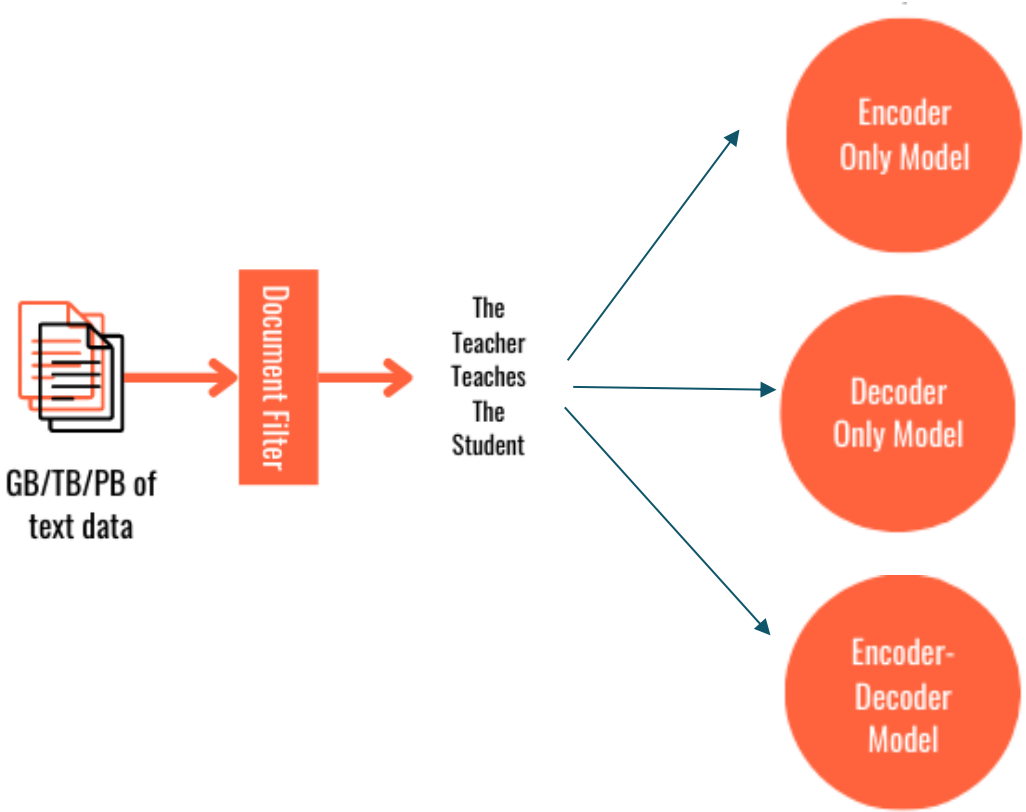
Here's how we can format the input:
[START] *The man is in the doorway* [DELIM] *The person is near the door* [EXTRACT]

Increasingly convincing generations (GPT2)

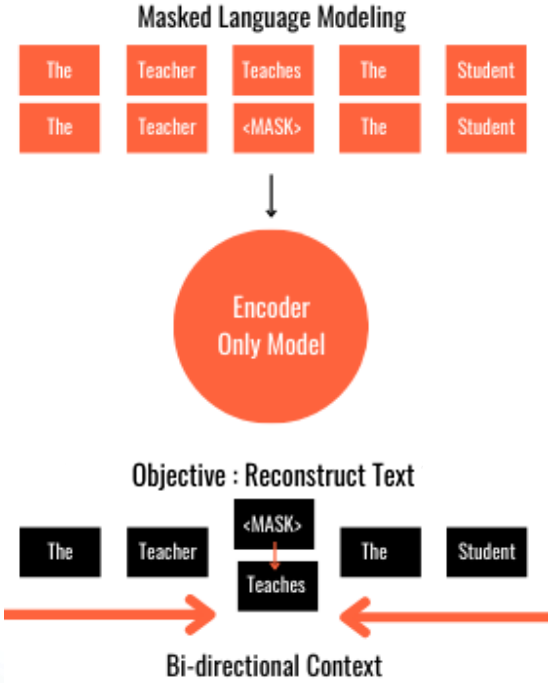
GPT-2, a larger version (1.5B) of GPT trained on more data, was shown to produce relatively convincing samples of natural language.

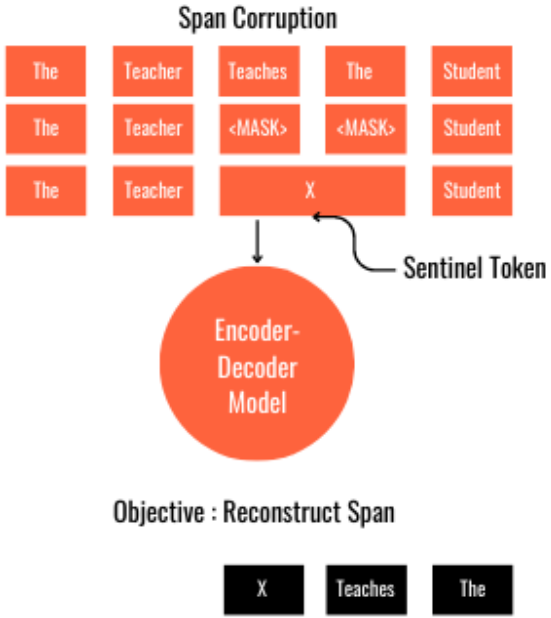
Context (human-written): In a shocking finding, scientist discovered a herd of unicorns living in a remote, previously unexplored valley, in the Andes Mountains. Even more surprising to the researchers was the fact that the unicorns spoke perfect English.
GPT-2: The scientist named the population, after their distinctive horn, Ovid's Unicorn. These four-horned, silver-white unicorns were previously unknown to science. Now, after almost two centuries, the mystery of what sparked this odd phenomenon is finally solved. Dr. Jorge Pérez, an evolutionary biologist from the University of La Paz, and several companions, were exploring the Andes Mountains when they found a small valley, with no other animals or humans. Pérez noticed that the valley had what appeared to be a natural fountain, surrounded by two peaks of rock and silver snow.

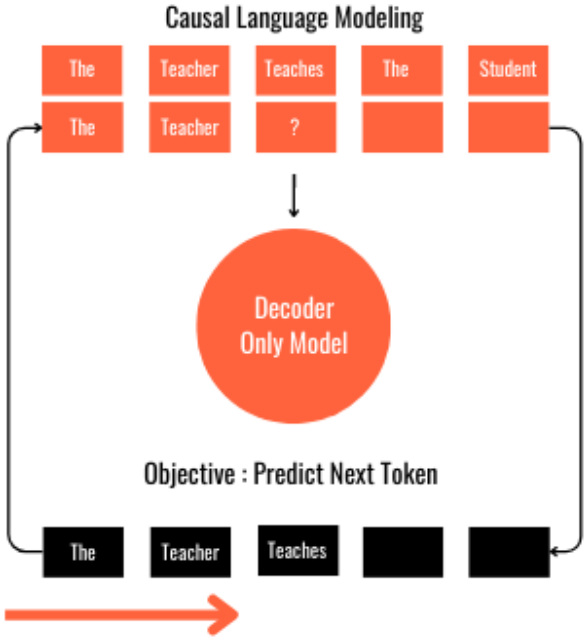
Let's put them all together



Let's put them all together

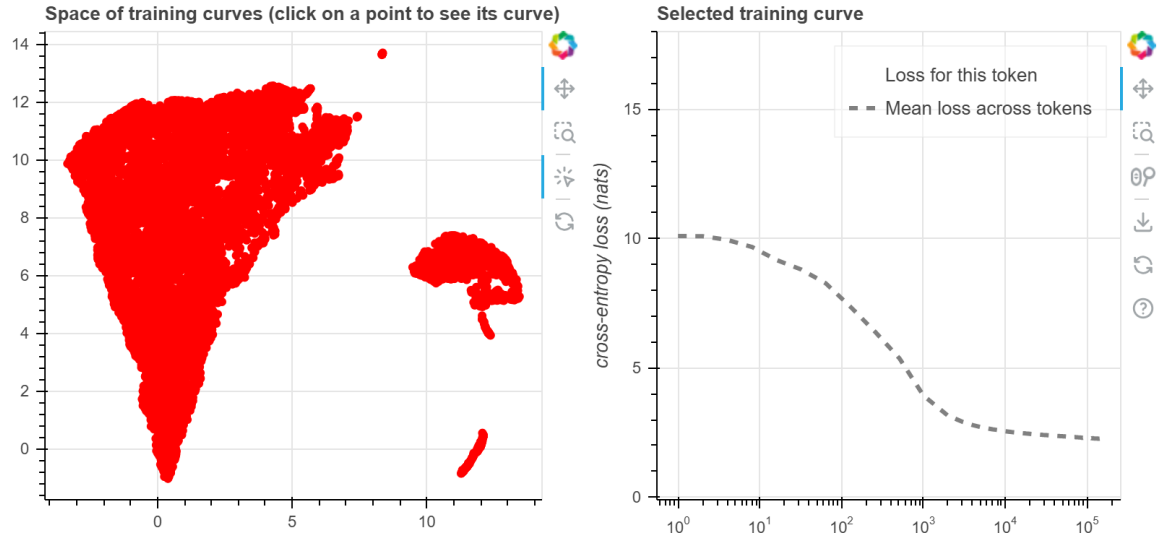






Visualizing LLM learning curves

An interactive visualization exploring how the loss of individual token predictions changes throughout LLM pretraining.



4.

Data – how to find and use it

Main Resource For this Section: https://huggingface.co/spaces/HuggingFaceFW/blogpost-fineweb-v1, https://commoncrawl.org/, [Paster et al., 2024], [Liu et al., 2024] and https://cmu-l3.github.io/anlp-spring2026/

Data factors

- **Quantity:** How much data do I have?
- **Quality:** Is it beneficial for training?
- **Coverage:** Does the data cover the domain(s) I care about?



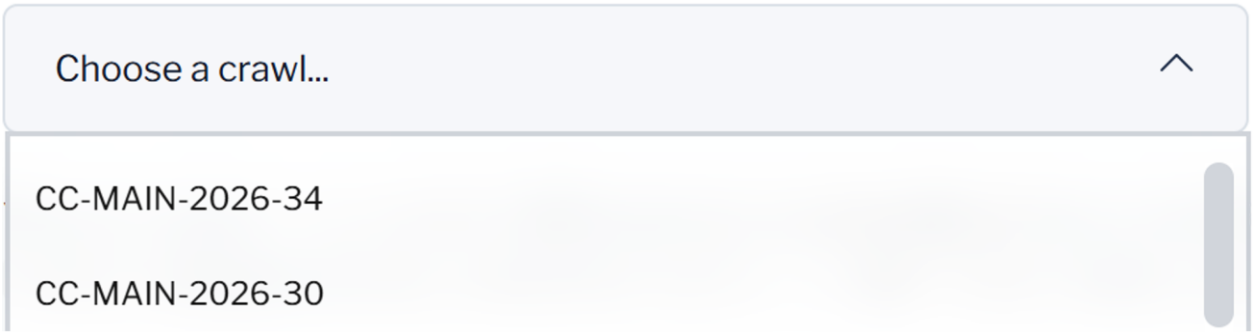
Quantity

	Tokens of training data
Llama 1	1.4 trillion
Llama 2	1.8 trillion
Llama 3	15 trillion
Deepseek 3	15 trillion

CommonCrawl

A non-profit organization that **crawls** the internet.
Contains petabytes of data, regularly collected since 2008.

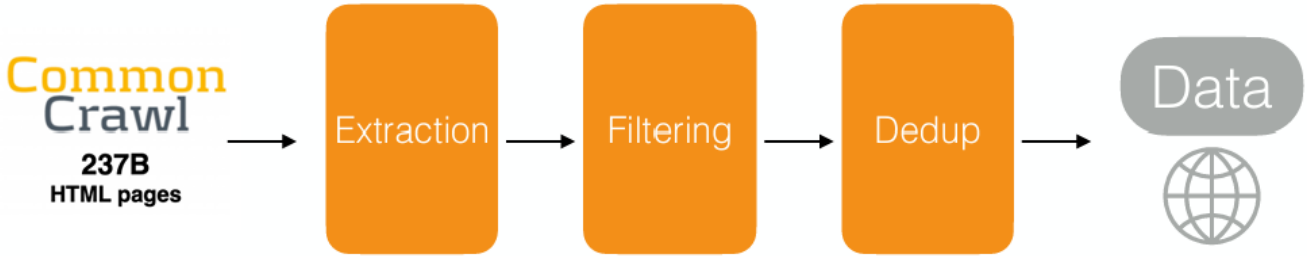
- **Scans the Web:** A specialized web crawler called **CCBot** scans billions of web pages every month.
- **Stores Archives:** It compiles these scans into massive datasets totaling petabytes of information. The corpus contains raw web page data and text extracts.
- **Hosts for Free:** The data is stored and made freely downloadable.

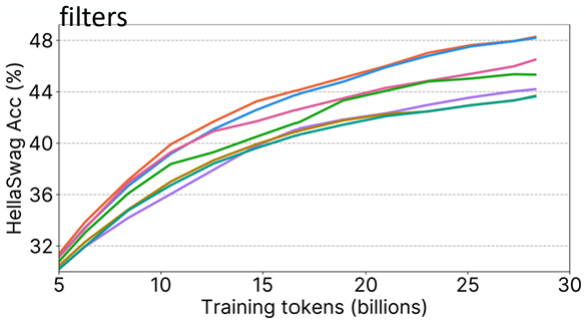


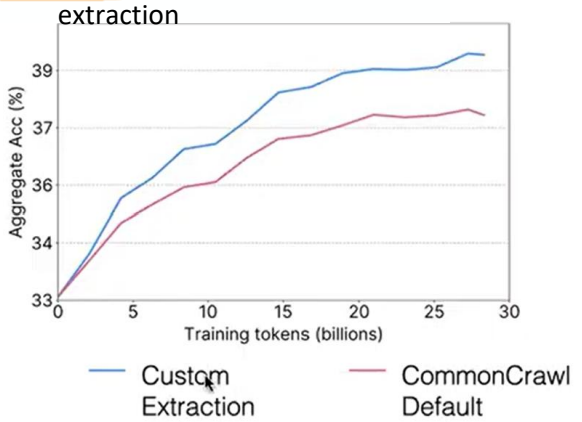


Quality

- **Extraction**: HTML to text
- **Filtering**: filter out unwanted pages
- **Deduplication**: many duplicate web pages





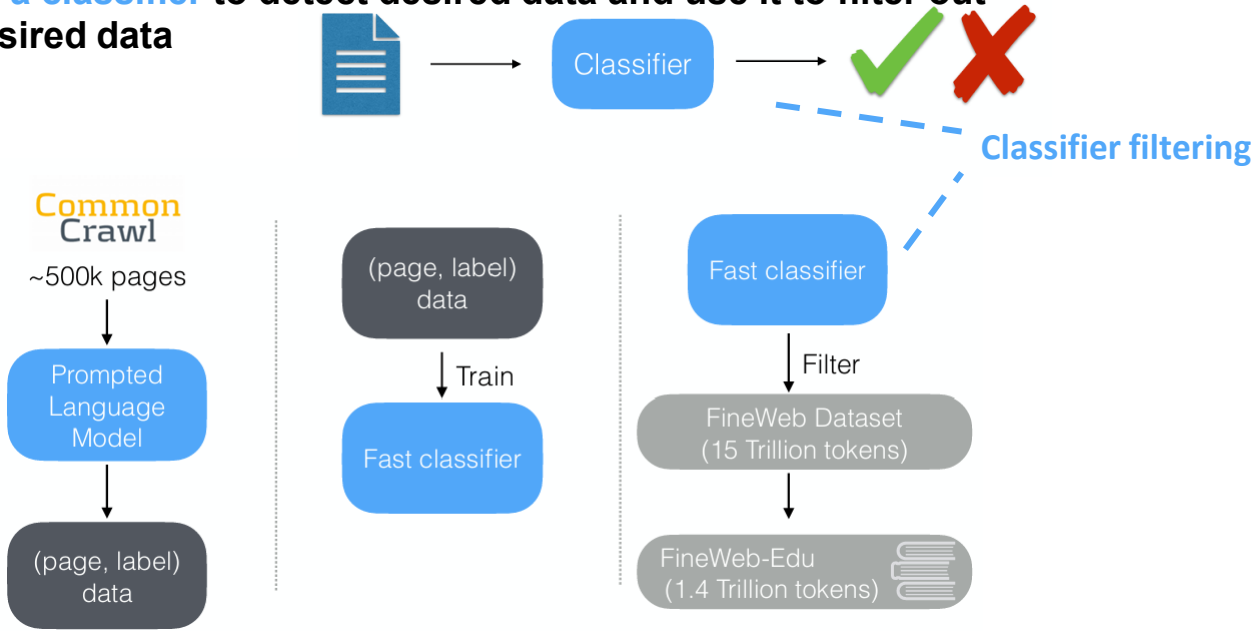


Coverage

Language models are trained to mimic the text distribution they are fed.

- Web data ≠ math data
- Web data ≠ educational data

Train a classifier to detect desired data and use it to filter out undesired data



Use a mixture of different sources

Having more training data is not enough. The composition of that data matters.

- For example, if we have 6 trillion tokens available for training, should we use 90% web pages? 50% web pages? Should we include more mathematical content or code?

Composition of Dolma 3 Mix

Source	Type	9T Pool		6T Mix	
		Tokens	Docs	Tokens	Docs
Common Crawl	Web pages	8.14T	9.67B	4.51T (76.1%)	3.15B
olmOCR science PDFs	Academic documents	972B	101M	805B (13.6%)	83.8M
Stack-Edu (Rebalanced)	GitHub code	137B	167M	409B (6.89%)	526M
arXiv	Papers with LaTeX	21.4B	3.95M	50.8B (0.86%)	9.10M
FineMath 3+	Math web pages	34.1B	21.4M	152B (2.56%)	95.5M
Wikipedia & Wikibooks	Encyclopedic	3.69B	6.67M	2.51B (0.04%)	4.24M
Total		**9.31T**	**9.97B**	**5.93T (100%)**	**3.87B**

Why Use a mixture of different sources?

Imagine we want to train a general-purpose LLM.
We could collect data from several sources:

Source	What the model might learn
Web pages	General knowledge and language patterns
Scientific papers	Technical and scientific terminology
Code	Programming patterns and syntax
Mathematical content	Mathematical notation and problem-solving patterns
Wikipedia	Encyclopedic knowledge

- If we train only on web pages, the model may have relatively less exposure to high-quality mathematical or scientific material.
- But we also cannot simply include equal amounts of everything.
- Different sources have different sizes, quality levels, and usefulness.

So we need to choose **a data mixture**.

How do we find the best mixture?

- **Step 1** — Classify documents into topics and quality levels
- **Step 2** — Train many small models with different mixtures
- **Step 3** — Fit a regression model
- **Step 4** — Optimize the mixture
- **Step 5** — Favor high-quality documents when upsampling



Classify documents and Train small models

First categorize the documents.

A document might receive labels such as:

- Topic: mathematics, science, code, general web, etc.
- Quality: high, medium, low.

This lets us control not only how much mathematics to include, but also which mathematical documents to prioritize.

Next step

Training a huge LLM for every possible mixture would be extremely expensive →Instead, train smaller models as experiments.

Small model	Web	Math	Code
A	80%	10%	10%
B	60%	30%	10%
C	50%	20%	30%

Discover which mixtures work better by evaluating the models on tasks such as mathematical problem solving and code generation.

Fit a regression model and Optimize the mixture

For each evaluation task, train a regression model to predict the performance of an LLM trained with that mixture.

$$\text{f(mixture)} \rightarrow \text{task performance} \quad \xrightarrow{\text{for mixture m} = [0.6, 0.3, 0.1]} \quad \begin{matrix} f_{math}(m) = 75 \\ f_{code}(m) = 62 \end{matrix}$$

60% web
30% math
10% code



Next step

A function that predicts performance from mixture percentages



Search for a mixture that performs well across evaluation tasks

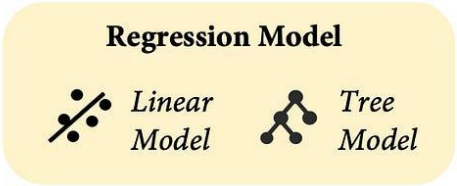
Example

1 Train **small-scale** proxy models

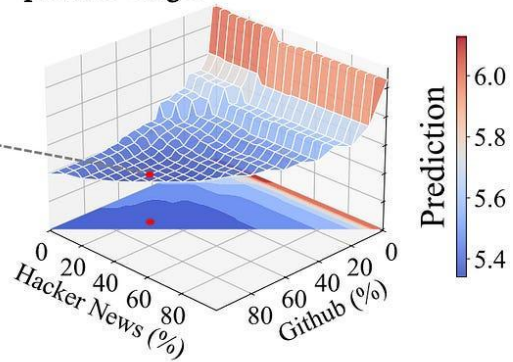
Hacker News	Github	Philpapers	Target ↓
9.5%	35.9%	54.6%	5.46
87.7%	12.0%	0.3%	5.57
24.4%	1.4%	74.2%	6.07
...			...



2 Fit a **regression** model using data mixture as source



3 **Simulate** new data mixtures and predict Target



4 Train a **large-scale** model on the best mixture

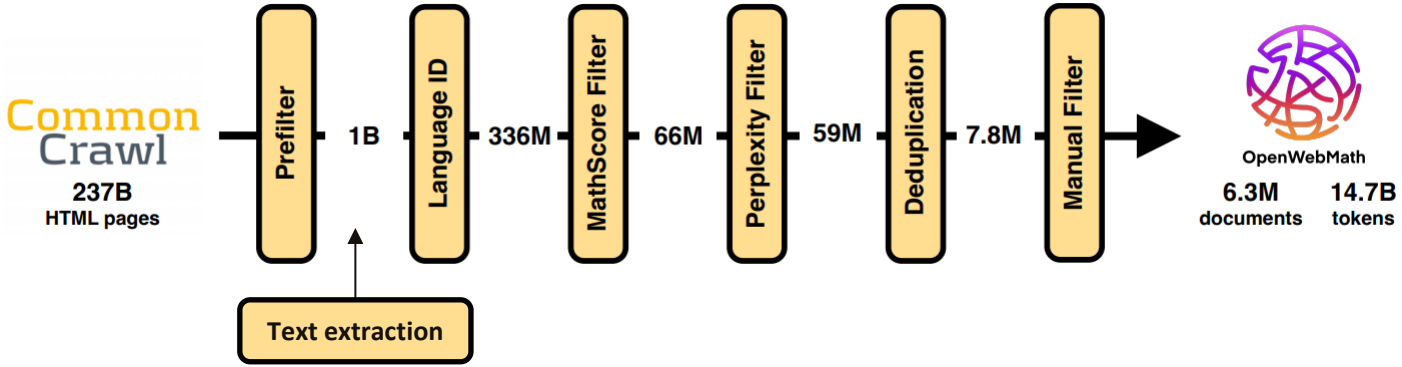
Simulated Best Data Mixture

Hacker News	Github	Philpapers
22.8%	67.0%	10.2%

Predicted Best Target

Prediction (Lowest)
5.34

OpenMathWeb



- Reduce the number of documents that need to be processed.
- Filtering with the goal of removing non-English documents
- Filtering with MathScore classifier: detects math content
- Perplexity filter
- Remove near-duplicate documents
- Manual filter

FineWeb

- Extraction
- URL filters (adult and blacklisted domains)
- Language filter: English only
- Quality filters
- Deduplication

Prompt LLM to “grade” rows of data in FindWeb, and filter them out

FineWeb-Edu

FineWeb-Edu

Below is an extract from a web page. Evaluate whether the page has a high educational value and could be useful in an educational setting for teaching from primary school to grade school levels using the additive 5-point scoring system described below. Points are accumulated based on the satisfaction of each criterion:

- Add 1 point if the extract provides some basic information relevant to educational topics, even if it includes some irrelevant or non-academic content like advertisements and promotional material.
- Add another point if the extract addresses certain elements pertinent to education but does not align closely with educational standards. It might mix educational content with non-educational material, offering a superficial overview of potentially useful topics, or presenting information in a disorganized manner and incoherent writing style.
- Award a third point if the extract is appropriate for educational use and introduces key concepts relevant to school curricula. It is coherent though it may not be comprehensive or could include some extraneous information. It may resemble an introductory section of a textbook or a basic tutorial that is suitable for learning but has notable limitations like treating concepts that are too complex for grade school students.
- Grant a fourth point if the extract highly relevant and beneficial for educational purposes for a level not higher than grade school, exhibiting a clear and consistent writing style. It could be similar to a chapter from a textbook or a tutorial, offering substantial educational content, including exercises and solutions, with minimal irrelevant information, and the concepts aren't too advanced for grade school students. The content is coherent, focused, and valuable for structured learning.
- Bestow a fifth point if the extract is outstanding in its educational value, perfectly suited for teaching either at primary school or grade school. It follows detailed reasoning, the writing style is easy to follow and offers profound and thorough insights into the subject matter, devoid of any non-educational or complex content.

The extract: <extract>.

After examining the extract:

LLM output

- Briefly justify your total score, up to 100 words.
- Conclude with the score using the format: "Educational score: <total points>"

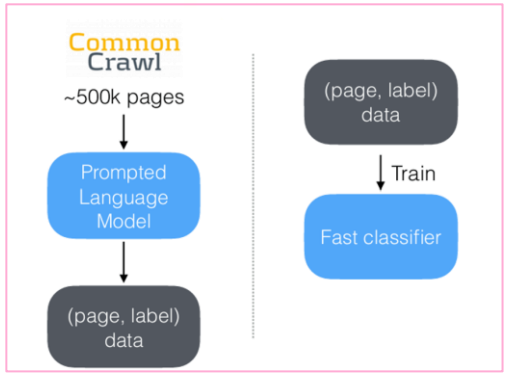
FineWeb-Edu

LLM output

The extract from the web page provides detailed information about the 2024 Enga landslide in Papua New Guinea, including its causes, impact, and aftermath. It addresses various educational elements such as geography, natural disasters, and humanitarian responses. The content is coherent, organized, and introduces relevant key concepts, making it suitable for educational use. However, it may be too complex for primary school students and includes some extraneous information not directly related to education. Therefore, it aligns well with grade school curricula but has limitations for lower education levels.

Educational score: 4 points

Too expensive! →



→ LLM-trained classifier

Converted the problem to a binary classification task by using a fixed threshold to determine if a file is educational(With a threshold of 3)

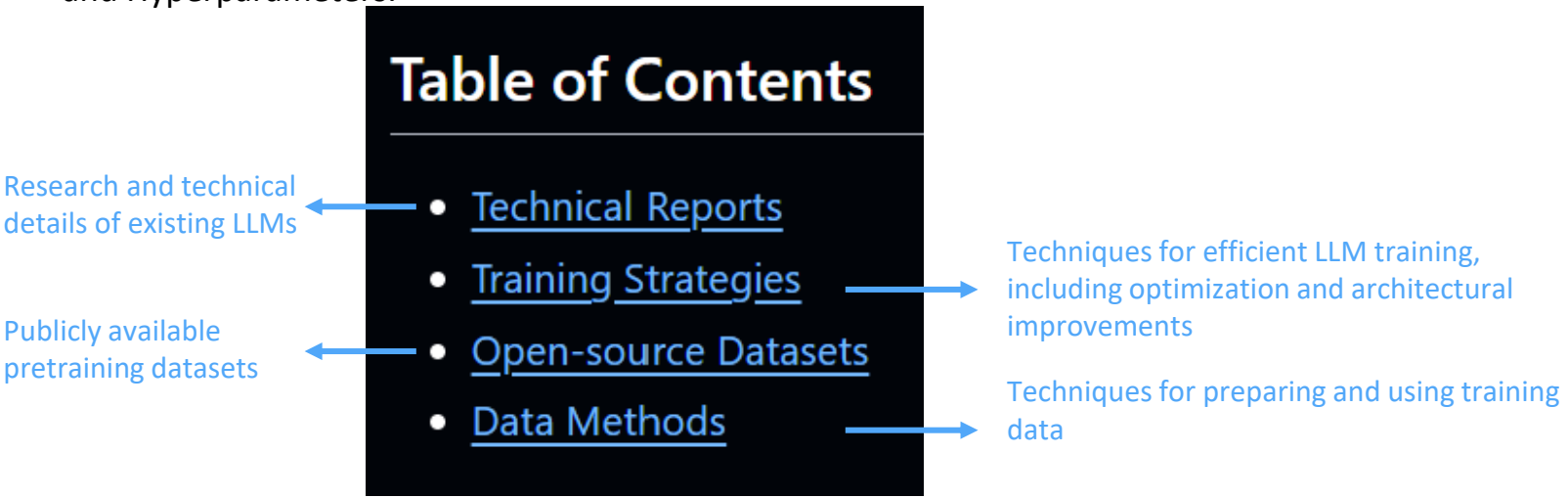
Other examples

	Year	Domain	Tokens
FineWeb	2024	Web	15 trillion
RedPajama v2	2024	Web	30 trillion
Dolma	2024	Mix	3 trillion
OLMO2 Mix	2025	Mix	4 trillion
DOLMA 3	2025	Mix	6 trillion
OpenWebMath	2023	Math web pages	15 billion
AlgebraicStack	2023	Math code	11 billion
FineWeb-Edu	2024	Educational (middle-school)	1.4 trillion

A collection of resources

A collection of papers, technical reports, datasets, and open-source tools related to large language model pretraining.

- Covers all major components of LLM pretraining: Architecture, Task, Data and Hyperparameters.



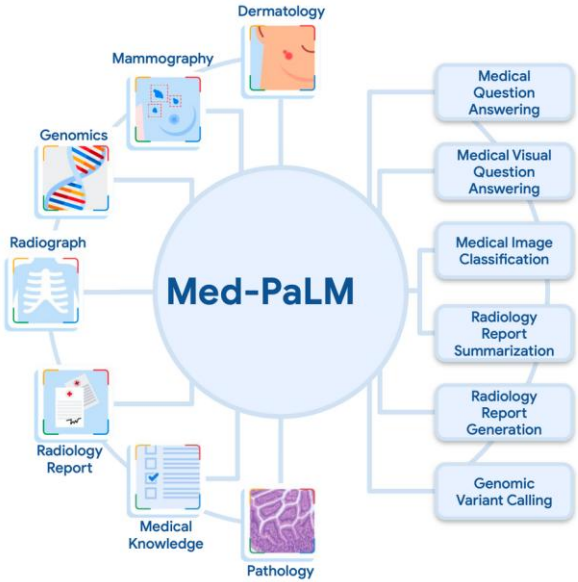
4.

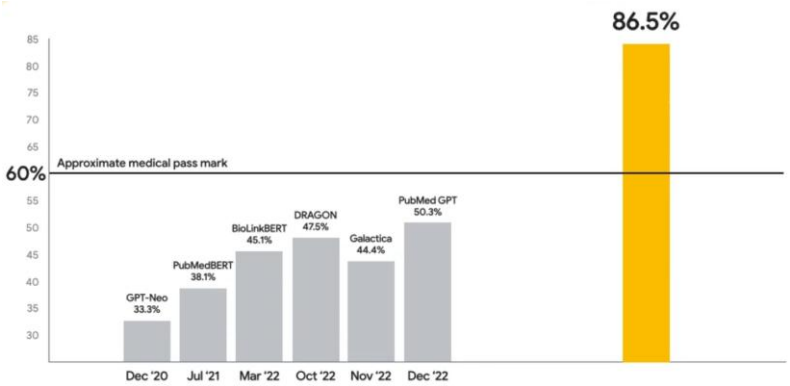
Specialized LLMs

Main Resource For this Section: https://sites.research.google/gr/med-palm/, https://www.finatune.com/en/ai/models/finance-specialized/harvey-llm,

Healthcare LLMs: Med-PaLM

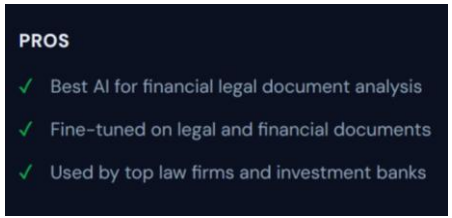
Healthcare LLMs are trained on medical literature, clinical records, and biomedical databases to understand medical terminology, reasoning patterns, and clinical workflows.
Starting with a pretrained general-purpose LLM (PaLM 2), Google further trained it on medical question-answering datasets to improve its performance on medical tasks.





Legal LLMs: Harvey AI

Legal LLMs process contracts, case law, and regulatory documents with domain-specific understanding of legal language and reasoning.
Harvey AI is fine-tuned on legal data. It handles contract analysis, legal research, due diligence, and regulatory compliance review. Used by major law firms including Allen & Overy.





Advanced NLP - Session 8 - LLM Pretraining: Data pipelines, Filtering and Curation - IUST

Intuition | Motivating word meaning and context | Architectures and their objectives | Data how to find and use it | Specialized LLMs